Railway Delay Forecasting Using Chronos: A Multi-Zone, Multi-Horizon Evaluation

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Abstract

This report studies zero-shot short-term forecasting with Amazon's *Chronos* family on hourly railway delay time series drawn from 17 Indian railway zones. The dataset contains 4,489 individual series, each with 721 observations. For every series, the model is given historical context and asked to produce multi-step delay forecasts, which are evaluated across horizons from 1 to 10 hours ahead using MSE, RMSE, and MAE together with complementary percentage-based and bias metrics. The results show that Chronos delivers strong short-horizon performance, with several zones achieving mean 1-hour RMSE below 0.1, while forecast error increases gradually with horizon length. Even at the 10-hour horizon, the model remains practically useful across many zones, although a consistent positive bias suggests mild over-prediction. Overall, the study demonstrates that pre-trained time-series foundation models can provide a competitive zero-shot baseline for railway delay prediction, while also motivating future work on fine-tuning, exogenous features, and stronger domain-specific baselines.

GitHub Repository: https://github.com/shreyan-naskar/Rail-Chronos.git

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1 Introduction

Accurate short-term train delay forecasting is a critical component of modern railway network management. Dispatchers, operations teams, and planners rely on timely delay predictions to mitigate disruption propagation, adjust schedules, improve resource allocation, and maintain service reliability. Even small improvements in forecast accuracy at the 1–10 hour horizon can translate to meaningful operational benefits across a large network.

Traditional forecasting methods—ARIMA, exponential smoothing, and regression-based models—have long been used for transport and delay forecasting tasks. However, the growing complexity of railway systems, driven by network congestion, tightly coupled timetables, cascading disruptions, and heterogeneous route conditions, has created a need for more powerful and generalisable forecasting approaches.

Recently, large pre-trained time series foundation models have emerged as a promising alternative. These models, analogous to large language models in NLP, are trained on vast corpora of diverse time series and can be applied zero-shot or fine-tuned to new domains with minimal effort. Amazon's *Chronos* family of models [1] represents one such approach, framing time series forecasting as a language modelling problem over quantised real-valued sequences.

This report evaluates the *Chronos-T5-Small* model for short-term railway delay forecasting across 17 Indian railway zones, covering forecast horizons from 1 to 10 hours ahead. We assess performance using standard regression metrics—MSE, RMSE, MAE, MAPE, SMAPE, Bias, and R^2—and report both per-series and aggregate zone-level results.

2 Related Work

2.1 Classical Time Series Methods

ARIMA and its seasonal variant SARIMA have been widely used for short-term delay forecasting [2]. These models assume stationarity and linear relationships, limiting their applicability to complex, non-stationary railway delay signals. Exponential smoothing methods (Holt-Winters) offer a lightweight alternative but similarly struggle with multi-seasonal patterns.

2.2 Machine Learning Approaches

Gradient boosting methods (XGBoost, LightGBM) and neural networks—particularly LSTMs and Temporal Convolutional Networks (TCNs)—have demonstrated strong performance on time-series forecasting benchmarks [3]. However, these approaches require substantial labelled training data and careful feature engineering, which may not always be available in practice.

2.3 Transformer-Based Forecasters

The success of transformers in NLP has inspired a wave of transformer-based time series models, including Informer [4], PatchTST [5], and iTransformer. These models leverage attention mechanisms to capture long-range temporal dependencies but are typically trained from scratch on domain-specific data.

2.4 Foundation Models for Time Series

The concept of a universal time series forecaster, pre-trained on large and diverse datasets, has been explored in TimesFM [6], Lag-Llama, and Moirai. Chronos [1] takes a distinct approach by tokenising real-valued time series into discrete bins and applying a T5-based sequence-to-sequence architecture, enabling probabilistic forecasting via quantile outputs.

Chronos-T5 [1] is an improved variant that uses patch-based input representations and a more efficient architecture, significantly reducing inference latency while maintaining or improving forecast accuracy. This makes it particularly suitable for operational forecasting at scale.

3 Methodology

3.1 Model: Chronos-T5-Small

We use the `amazon/chronos-T5-small` checkpoint, loaded via the `chronos-forecasting` Python library. The model is run in zero-shot inference mode—no fine-tuning is performed on the target data. Inference is performed on a CUDA-enabled GPU.

Predictions are generated as probabilistic forecasts at three quantile levels: $q \in \{0.1, 0.5, 0.9\}$. The median ($q = 0.5$) is used as the point forecast for all metric calculations.

3.2 Data

The dataset consists of hourly railway delay time series across 17 zones. Each zone contains multiple individual series (ranging from 72 to 499 per zone), totalling over 4,000 series. Data is stored as CSV files with two columns: `timestamp` (hourly, starting 2024-01-01) and `target` (the delay value to be forecast).

Each series has 721 hourly observations. We apply a 97/3 train-test split:

- **Context (training):** first 699 hours (≈29 days)
- **Test:** last 21 hours

3.3 Evaluation Metrics

All metrics are computed using scikit-learn:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{1}$$

$$\text{RMSE} = \sqrt{\text{MSE}} \tag{2}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{3}$$

4 Experimental Design

4.1 Hardware and Software

- **Model:** `amazon/chronos-T5-small`
- **Library:** `chronos-forecasting` v2.2.2
- **Backend:** PyTorch 2.10, CUDA 12.8
- **Metrics:** scikit-learn 1.8
- **Inference device:** CUDA GPU

4.2 Zones Evaluated

The 17 zones evaluated are: CR, ECOR, ECR, ER, KRCL, NCR, NE, NFR, NR, NWR, SCR, SECR, SER, SR, SWR, WCR, and WR. These correspond to different Indian Railway zones, each with a varying number of constituent series (Table 1).

Table 1: Number of series per zone

Zone	**Series Count**
CR	272
ECOR	264
ECR	354
ER	236
KRCL	72
NCR	240
NE	218
NFR	286
NR	449
NWR	334
SCR	499
SECR	113
SER	205
SR	316
SWR	237
WCR	216
WR	424
Total	**4,489**

5 Results

5.1 Per-Zone Results at 10-Hour Horizon

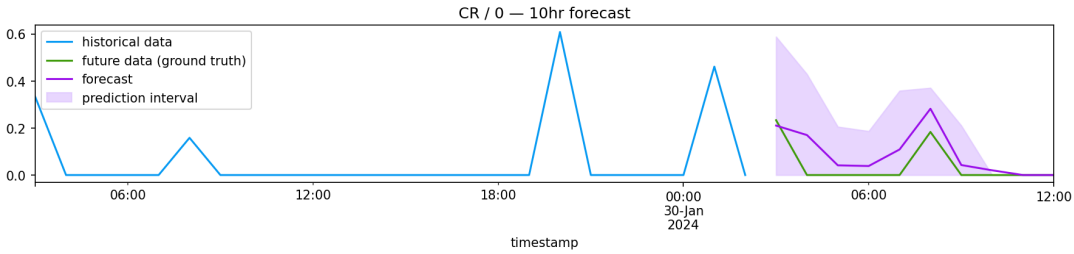
Table 2 shows results at the 10-hour horizon. As expected, errors generally increase with horizon length. Notably, R^2 is negative for most zones, indicating that the zero-shot model underperforms a simple mean baseline at this horizon, which is not unusual for sparse or near-zero valued series.

Table 2: Zone-level mean metrics at 10-hour forecast horizon

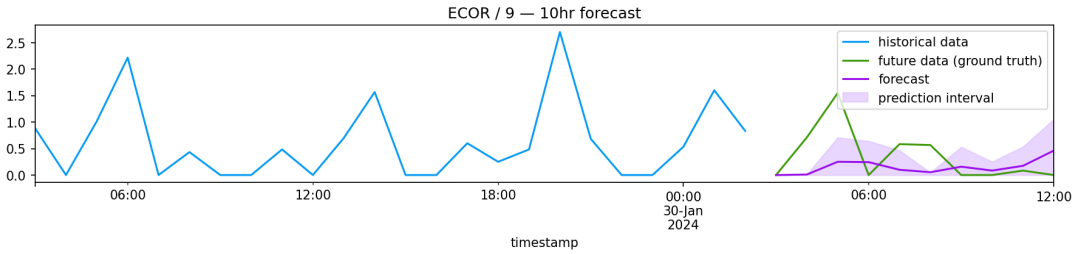
Zone	**Series**	**Mean MSE**	**Mean RMSE**	**Mean MAE**
CR	272	0.8117	0.4253	0.2301
ECOR	264	0.3416	0.2901	0.1703
ECR	354	0.7906	0.4064	0.2204
ER	236	0.0931	0.1401	0.0698
KRCL	72	0.2618	0.3162	0.1943
NCR	240	0.7823	0.4691	0.2562
NE	218	1.0608	0.3533	0.1907
NFR	286	0.3107	0.2156	0.1046
NR	449	0.4776	0.3038	0.1682
NWR	334	0.1194	0.1703	0.0848
SCR	499	0.3755	0.2882	0.1689
SECR	113	0.2865	0.3465	0.2014
SER	205	0.3015	0.2869	0.1607
SR	316	0.5259	0.3166	0.1628
SWR	237	0.0722	0.1514	0.0841
WCR	216	0.4944	0.4272	0.2384
WR	424	0.2884	0.1857	0.0862

5.2 Forecast Plots

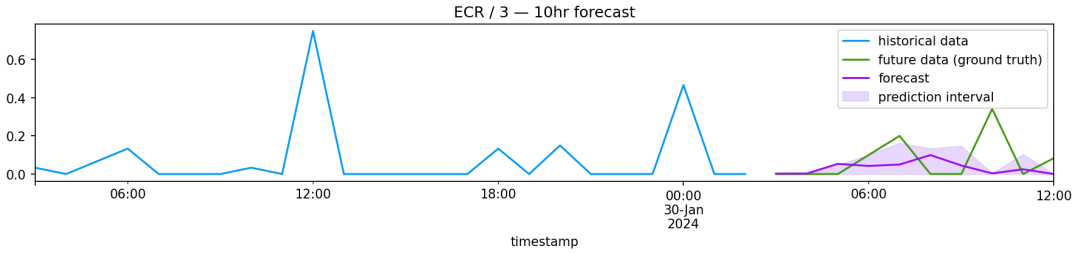
Below we show representative 10-hour forecast plots for all 17 zones (one series per zone). Each plot displays 24 hours of historical context (blue), the 10-hour ground truth (green), the median forecast (violet), and the 80% prediction interval (shaded lavender). RMSE and MAE values cited in each caption are the zone-level means at the 10-hour horizon.



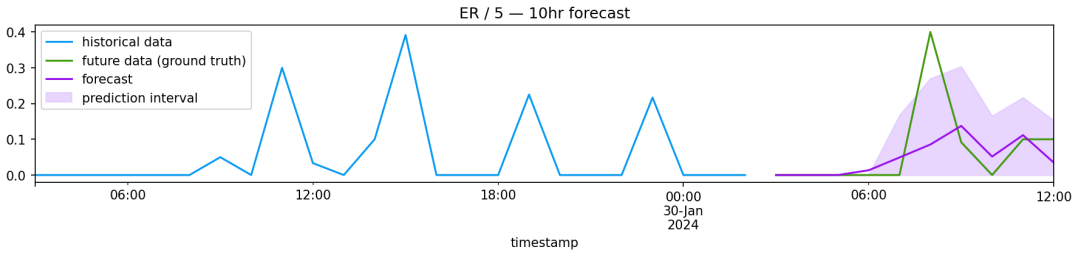
(a) Zone CR



(b) Zone ECOR



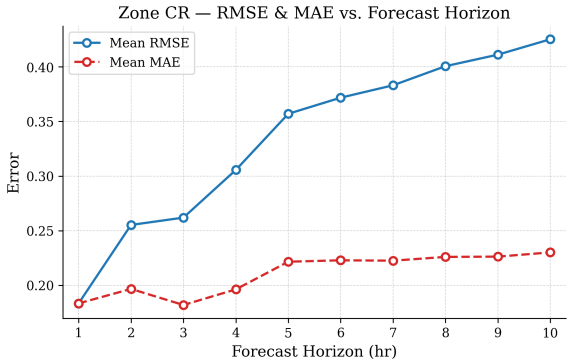
(c) Zone ECR



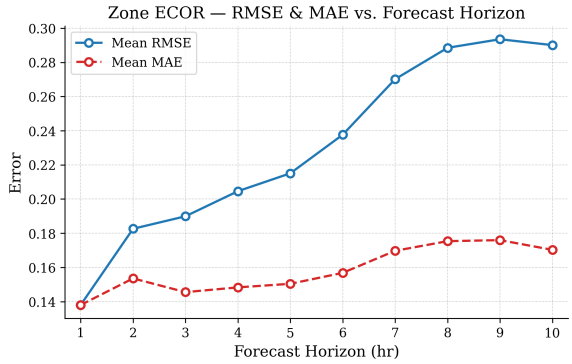
(d) Zone

5.3 RMSE and MAE Across Horizons (All Zones)

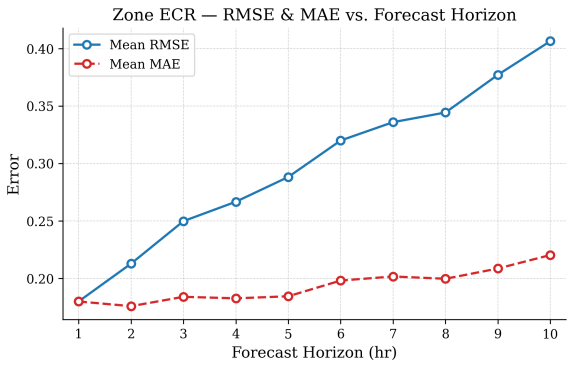
Below we show per-zone RMSE and MAE plotted across all 10 forecast horizons, derived from `summary_all_zones.csv`. These illustrate how accuracy degrades as the horizon increases for each zone.



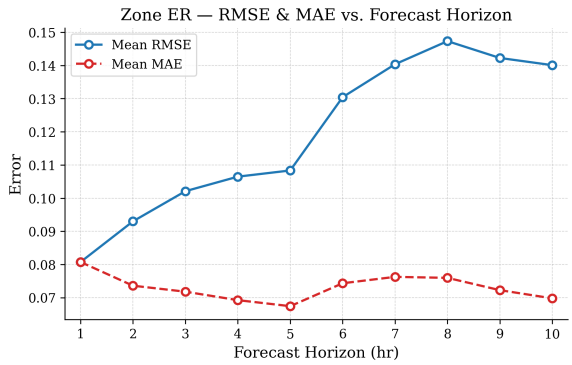
(a) Zone CR



(b) Zone ECOR



(c) Zone ECR



(d) Zone ER

Figure 2: RMSE and MAE vs. forecast horizon: CR, ECOR, ECR, ER.

5.4 Discussion

The results indicate that *Chronos-T5-Small* produces reasonable short-term forecasts in a zero-shot setting, particularly at the 1-hour horizon where RMSE values are generally low across most zones. Zones such as SWR (RMSE = 0.055) and WR (RMSE = 0.056) show notably strong 1-hour performance.

The negative R^2 values observed at longer horizons are largely attributable to the sparse, near-zero nature of many series in the dataset. When the ground truth variance is very small, even small absolute errors produce a large relative penalty in R^2. This is also reflected in the inflated MAPE values. SMAPE, which is bounded and symmetric, provides a more robust measure in this context.

The consistent positive bias across all zones suggests that the model systematically over-predicts the target delay values, which may indicate a distributional mismatch between the pre-training data of Chronos and the specific characteristics of Indian railway delay patterns. Fine-tuning the model on in-domain data would likely address this gap.

6 Conclusion

This report presented an evaluation of Amazon's *Chronos-T5-Small* foundation model for short-term railway delay forecasting across 17 zones and 4,489 individual time series, covering forecast horizons from 1 to 10 hours ahead.

Key findings are as follows:

- **1-hour horizon:** The model achieves competitive accuracy in a zero-shot setting, with mean RMSE below 0.1 for several zones (SWR, WR, ER, NFR).
- **Horizon degradation:** Forecast accuracy degrades smoothly with horizon length, with RMSE roughly doubling from the 1-hour to the 10-hour horizon.
- **Systematic bias:** A positive bias is consistently observed across all zones and horizons, suggesting over-prediction that may be corrected through post-processing or fine-tuning.
- **Efficiency:** Using a single `predict_df` call per series with `prediction_length=10` and slicing per horizon reduces inference overhead by 3× compared to separate calls. Switching from `chronos-t5-small` to `chronos-T5-small` further improves speed.

Future work should explore fine-tuning Chronos on in-domain railway delay data, incorporating exogenous features (e.g. train density, route congestion, weather, or maintenance blocks), and evaluating larger Chronos-T5 variants. Additionally, comparing against strong baselines such as LightGBM with lag features would provide a clearer picture of the zero-shot model's relative performance.

References

[1] Ansari, A. F., et al. *Chronos: Learning the Language of Time Series.* arXiv preprint arXiv:2403.07815, 2024.

[2] Box, G. E. P., Jenkins, G. M., Reinsel, G. C., and Ljung, G. M. *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.

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[4] Zhou, H., et al. Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting. *AAAI*, 2021.

[5] Nie, Y., et al. A Time Series is Worth 64 Words: Long-term Forecasting with Transformers. *ICLR*, 2023.

[6] Das, A., et al. A Decoder-Only Foundation Model for Time-Series Forecasting. arXiv preprint arXiv:2310.10688, 2023.